

Defective Analysis

Formal Exploration of Geometry

Typed carrier · projector sector · quotient · residual fiber · admissible reconstruction

Geometry is not primitive space. Geometry is the admissible reconstruction of relational defect over a typed carrier.

Geometry = residual-fibered reconstruction of position, incidence, continuity, metric, curvature, and phase.

In matured Defective Analysis, no geometry is untyped. A geometric object is specified relative to a carrier, coefficients, witness data, equivalence, projectors, quotient layer, retained kernel, and residual comparator.

$G = (X, A, W, E, \Pi, Q, K, R_{\text{cmp}})$

The quotient Q is the exposed geometric layer. The retained kernel K is the hidden fiber required to reconstruct what the quotient killed.

$D_{\text{theta}}(X) = (Q, K)$
 $I_{\text{alpha}}(Q, K) \sim_E X$
 $I_{\text{alpha}}(Q, \emptyset) \not\sim_E X \Rightarrow K \text{ is geometrically real.}$

0. Terminal claim

Geometry is not the assumption of a pre-existing ambient space. It is the stabilized reconstruction of relations among distinguishable witnesses after quotienting and residual retention.

A mature DA geometry must therefore state its type, carrier, coefficient structure, projectors, quotient, residual fiber, and admissibility condition.

```
G = (X, A, W, E, Pi, Q, K, R_cmp)
D_theta X = (Q, K)
I_alpha(Q, K) ~_E X
```

1. Geometry from the algebraic ladder

The arithmetic ladder repairs operations. The geometric ladder spatializes those repairs as relational structure.

The same ascent from 0 to $\mathbb{C}$ induces a geometry ladder from localization to phase geometry.

```
0 -> N -> Z -> Q -> R -> C
0 -> Pt -> Ray_N -> Line_Z -> Aff_Q -> Geom_R -> PhaseGeom_C

0 = no exposed distinction
Pt = first localized distinction
Ray_N = forward discrete extension
Line_Z = oriented reversible extension
Aff_Q = ratio/intersection geometry
Geom_R = continuous metric/smooth geometry
PhaseGeom_C = rotation, phase, holomorphic closure
```

2. The point

A point is not an atom. A point is a stabilized equivalence class of local witnesses.

The point is the first geometric repair of the zero class. Pure zero has no exposed localization; the point repairs indistinguishability by exposing location.

```
p = [w]_E, with w in W
I_alpha(D_theta(p)) ~_E p
0 ~> Pt
```

Known unknown: whether DA should treat points as primitive witnesses, quotient classes of witnesses, or terminal reconstruction residues.

Unknown unknown: whether there are geometric regimes where “point” is always a bad chart and must be replaced by region, fiber, event, or process.

3. Incidence geometry

Given a coefficient structure A , define the affine A -module $V_A = A^n$. A line through p in direction v is the orbit obtained by scalar extension along A .

Incidence asks which objects meet. It does not yet ask distance, angle, smoothness, or curvature.

For $A = \mathbb{N}$, incidence is forward-only. For $A = \mathbb{Z}$, it is reversible. For $A = \mathbb{Q}$, it admits rational interpolation and slope. For $A = \mathbb{R}$, it becomes continuum-stable. For $A = \mathbb{C}$, it carries phase and rotation.

```
V_A = A^n
L(p, v) = { p + t v : t in A }
p in L
Incidence geometry = algebraic closure made spatial.
```

Known unknown: whether incidence should be treated as the first geometric relation or whether adjacency/preorder comes earlier.

Unknown unknown: whether certain DA geometries admit incidence without points, using only morphisms or overlaps.

4. Affine geometry

Affine geometry appears when translation is admissible. It preserves collinearity and ratios but not length.

In DA terms, affine geometry is geometry modulo origin choice. The origin is a gauge choice. The difference vector is retained, while absolute position is quotient-relative.

Affine geometry is already residual-fibered: the exposed quotient is relative position data, and the killed or retained fiber is origin choice.

```
Aff(A^n) = A^n semidirect GL_n(A)
x -> Mx + b, with M in GL_n(A), b in A^n
v = q - p
Q = relative position data
K = discarded origin fiber
```

Known unknown: whether the affine origin should be treated as a killed gauge fiber or as a retained reconstruction kernel.

Unknown unknown: whether DA naturally prefers affine geometry over vector geometry because absolute origin is usually non-observable.

5. Metric geometry

A metric is a residual comparator. Distance is not primitive separation; it is the cost of reconstructing one witness from another under a selected chart.

Metric geometry requires $\mathbb{R}$ because limit-stable comparison requires completion. Without real completion, Cauchy-stable paths may have missing endpoints.

```
d : X × X -> R_{>=0}
d(x, y) = 0 <=> x ~_E y
d(x, y) = d(y, x)
d(x, z) <= d(x, y) + d(y, z)

d(x, y) = cost of reconstructing x from y under the chosen chart.
(x_n) Cauchy but lim x_n notin X => metric defect
X = Comp(X, d)
```

Known unknown: whether every DA residual comparator should be metric, pseudometric, asymmetric cost, divergence, or preorder.

Unknown unknown: whether physical distance is a special case of reconstruction cost or whether reconstruction cost is only an epistemic proxy.

6. Topological geometry

A topology is the grammar of admissible neighborhoods. In DA, an open set is not merely a container; it is a perturbation-stable reconstruction region.

Continuity means that admissible output neighborhoods pull back to admissible input neighborhoods. It prevents small accepted input defects from becoming inadmissible output defects.

```
tau subset P(X)
empty, X in tau
union_i U_i in tau
U, V in tau => U cap V in tau
```

```
f : X -> Y is continuous iff  $V \in \tau_Y \Rightarrow f^{-1}(V) \in \tau_X$ 
f continuous  $\Leftrightarrow$  f does not convert small admissible input defects into inadmissible
output defects.
```

Known unknown: whether topology should be derived from residual comparison or admitted independently as a primitive stability grammar.

Unknown unknown: whether DA supports non-classical topologies where neighborhoods are determined by projector sectors rather than subsets.

7. Smooth geometry

A smooth manifold is a multi-chart reconstruction object. Smoothness means the derivative clock survives chart transition.

A singularity appears where no admissible chart absorbs the residual. The failure may belong to the chart, quotient, projector, smooth structure, or retained kernel.

```
phi_i : U_i ->  $\mathbb{R}^n$ 
phi_j o phi_i^{-1} is smooth where defined
M = (|M|, tau, {phi_i}, E, Pi, K)

D_theta(phi_j o phi_i^{-1}) exists and remains admissible.
p in Sing(M)  $\Leftrightarrow$  for every admissible chart phi_i, local reconstruction near p fails.
```

Known unknown: whether DA singularities should be classified by failed chart, failed projector, failed quotient, or failed retained kernel.

Unknown unknown: whether some physical singularities are artifacts of forcing smooth charts where DA would use a different admissible category.

8. Tangent structure

The tangent space at a point is the first-order defect sector of curves through that point. It is the local grammar of possible departures.

The tangent bundle is therefore the geometry of possible local defects over the base space.

```
gamma : (-epsilon, epsilon) -> M
gamma(0) = p
v = gamma_dot(0) in T_pM
T_pM = space of admissible first-order departures from p
TM = disjoint_union_{pinM} T_pM
```

Known unknown: whether tangent vectors should be reconstructed from curves, derivations, germs, infinitesimal probes, or sectorized witness changes.

Unknown unknown: whether DA admits tangent structures in non-smooth settings without forcing classical differentiability.

9. Curvature

Curvature is the obstruction to path-independent reconstruction. It is the residual produced by transporting local reconstruction around a loop.

Flatness means loop reconstruction is path-independent. Curvature means the loop returns with residual. Thus curvature is failed commutation of local reconstruction operators.

This is the geometric version of the DA defect principle: derivative and integral clocks do not generally commute.

```
R_nabla(X,Y)Z = nabla_X nabla_Y Z - nabla_Y nabla_X Z - nabla_{[X,Y]} Z
R_nabla = [nabla_X, nabla_Y] - nabla_{[X,Y]}

R_nabla = 0 => flat reconstruction
R_nabla != 0 => loop residual
D_theta I_alpha != I_alpha D_theta
```

Known unknown: whether DA curvature is best identified with connection curvature, quotient curvature, holonomy residual, or projector noncommutativity.

Unknown unknown: whether all curvature-like effects in DA can be unified as failure of operation order invariance.

10. Geodesics

A geodesic is a locally self-reconstructing path. In connection language, it introduces no avoidable acceleration defect.

In metric geometry, geodesics locally extremize length. In DA geometry, geodesics minimize reconstruction correction under the chosen connection.

```
nabla_{\gamma_{dot}} \gamma_{dot} = 0
\gamma \text{ geodesic} \iff \text{local continuation introduces no avoidable acceleration defect.}
```

Known unknown: whether DA should define geodesics variationally, connection-theoretically, or through minimum residual accumulation.

Unknown unknown: whether some systems have multiple inequivalent geodesic notions depending on projector sector.

11. Fiber geometry

A fiber bundle separates exposed base structure from retained local freedom. In DA terms, the base is quotient structure and the fiber is retained kernel.

Gauge freedom is not meaningless redundancy. It is retained kernel structure invisible to the quotient but necessary for reconstruction.

```
pi : E -> B
B = base
F_b = pi^{-1}(b) = fiber over b
Q = B
K = F

s : B -> E, pi o s = id_B
A gauge choice is a section or local trivialization.
```

Known unknown: whether all DA retained kernels should be modeled as fibers.

Unknown unknown: whether some retained kernels are not bundle-like and require stacks, sheaves, higher categories, or nonlocal memory objects.

12. Gauge curvature

Given a gauge connection A , its curvature measures the defect of local fiber alignment under transport.

If the curvature vanishes, the gauge field is locally pure reconstruction choice. If it does not vanish, the retained fiber has observable residual.

Physics enters geometry when retained reconstruction fibers become dynamically constrained.

```
F_A = dA + A ^ A
F_A = defect of local fiber alignment under transport
F_A = 0 => locally pure reconstruction choice
F_A != 0 => observable residual
```

Known unknown: whether DA gauge curvature should be treated as fundamental or as a special case of projector-sector holonomy.

Unknown unknown: whether DA can classify physical interactions by the kind of retained kernel their quotient observables require.

13. Complex geometry

Complex geometry appears when the real line is insufficient to reconstruct phase. The complex coordinate retains real magnitude and phase closure.

A holomorphic function respects the complex phase structure. It does not treat x and y as independent real axes after the phase fiber has been admitted.

```
C = R[i] / (i^2 + 1)
z = x + i y
z -> e^{i\theta} z

dbar f = 0
C-geometry = R-geometry with retained phase closure.
```

Known unknown: whether complex structure should be treated as phase closure, orientation closure, or the minimal algebraic repair of real curvature defects.

Unknown unknown: whether DA naturally derives quantum phase from complex geometry or whether complex geometry is one representation of a deeper phase fiber.

14. Singularities

A singularity is not an infinite value. A singularity is a failed admissible reconstruction site.

A mature DA analysis must not collapse all singularities into one untyped failure. Each singularity must be typed by the sector that fails.

```
p in Sing(X) <=> no available chart makes p reconstructible with finite typed residual

Sing_chart = coordinate failure
Sing_metric = distance/completion failure
Sing_smooth = derivative failure
Sing_fiber = retained kernel failure
Sing_quotient = COTG overcompression failure
```

Known unknown: whether singularities should be repaired by completion, resolution, blowup, category change, or retained-kernel restoration.

Unknown unknown: whether physical singularities represent real ontic breakdowns or inadmissible quotient compression.

15. COTG as geometric quotient

For a geometric object X , the WDS contains more than visible geometry. It includes local witness data: charts, fibers, transitions, residuals, clocks, and projector sectors.

The COTG layer is the exposed curvature-parity quotient. It is not identical with the WDS. The lost structure is the kernel of the WDS-to-COTG projection.

Mature DA geometry is therefore always exposed quotient plus retained fiber plus residual comparison.

```
W_X = local witness data: charts, fibers, transitions, residuals, clocks, projector
sectors
COTG(X) = Pi(W_X) / E
COTG(X) != W_X
K_X = ker(W_X -> COTG(X))

I_alpha(COTG(X), K_X) ~_E X
```

16. Full DA geometry ladder

The completed DA geometry ladder can be read as a sequence of typed defect repairs. Each stage admits a new class of relational residuals and then reconstructs them as structure.

$\emptyset \rightarrow \text{Pt} \rightarrow \text{Inc} \rightarrow \text{Aff} \rightarrow \text{Met} \rightarrow \text{Top} \rightarrow \text{Diff} \rightarrow \text{Curv} \rightarrow \text{Fib} \rightarrow \text{Gauge} \rightarrow \text{Phase}$

$\emptyset$ = no exposed distinction

Pt = localized distinction

Inc = incidence

Aff = relative position

Met = reconstruction cost

Top = perturbation-stable neighborhood structure

Diff = stable derivative clock across charts

Curv = noncommuting reconstruction around loops

Fib = retained kernel over quotient base

Gauge = fiber-choice symmetry with observable curvature residual

Phase = complex/polar closure

Geometry is what arithmetic becomes when defects between elements must be reconstructed relationally.

$G = \text{Adm_DA}(X; A, \theta, \alpha, \Pi, E, Q, K, R_{\text{cmp}})$

$D_{\theta} X = (Q, K)$

$I_{\alpha}(Q, K) \sim_E X$

$I_{\alpha}(Q, \emptyset) \not\sim_E X \Rightarrow K \text{ is geometrically real.}$

Geometry is the residual fiber of failed flat reconstruction.